Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Equation of asymptotes of the hyperbola $xy - 3x - 4y + 8 = 0$, is

(A) $x = 3, y = 4$

(B) $x = 0, y = 0$

(C) $x = 4, y = 3$

(D) none of these

Q.2

The tangents to the hyperbola $x^2 - y^2 = 3$ are parallel to the straight line $2x + y + 8 = 0$ at the following points

(A) $(2, 1)$ or $(1, 2)$

(B) $(2, -1)$ or $(-2, 1)$

(C) $(-1, -2)$

(D) $(-2, -1)$

Q.3

The equation $16x^2 - 3y^2 - 32x - 12y - 44 = 0$ represents a hyperbola

(A) the length of whose transverse axis is $2\sqrt{3}$

(B) the length of whose conjugate axis is 8

(C) whose centre is at $(1, -2)$

(D) all of these

Q.4

The locus of point of intersection of perpendicular tangents of the hyperbola $4y^2 = x^2 - 1$, is

(A) $4x^2 + 4y^2 - 3 = 0$

(B) $2x^2 + 2y^2 - 5 = 0$

(C) $3x^2 + 3y^2 - 4 = 0$

(D) none of these

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Q.5

P is a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ transverse axis. The tangent to the hyperbola at P meets the transverse axis at T. If O is the centre of the hyperbola, then $OT.ON$ is equal to

(A) e^2

(B) a^2

(C) b^2

(D) b^2/a^2

Q.6

Two tangents are drawn to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is c^2. the locus of their point of intersection is

(A) $x^2-a^2=c^2\left(y^2+b^2\right)$

(B) $x^2+a^2=c^2\left(y^2-b^2\right)$

(C) $y^2+b^2=c^2\left(x^2-a^2\right)$

(D) none of these

Q.7

If $\frac{x^2}{k}+\frac{y^2}{k^2}=1(k<0)$ represents a hyperbola, then its

(A) transverse axis is x-axis

(B) transverse axis is y-axis

(C) transverse axis is neither x-axis nor y-axis

(D) conjugate axis is y-axis

Q.8

The equation of the hyperbola with vertices $(3,0)$ and $(-3,0)$ and semi latus rectum 4, is given by

(A) $4x^2-3y^2+36=0$

(B) $4x^2-3y^2+12=0$

(C) $4x^2-3y^2-36=0$

(D) none of these

Q.9

A hyperbola having transverse axis of length $2\sin\theta$ is confocal with the ellipse. $3x^2+4y^2=12$, then its equation is

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(A) $x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$

(B) $x^2 \sec^2 \theta - y^2 \operatorname{cosec}^2 \theta = 1$

(C) $x^2 \sin^2 \theta - y^2 \cos^2 \theta = 1$

(D) $x^2 \cos^2 \theta - y^2 \sin^2 \theta = 1$

Q.10

If PQ is a double ordinate of the hyperbola $\frac{\mathrm{x}^2}{\mathrm{a}^2} - \frac{\mathrm{y}^2}{\mathrm{b}^2} = 1$ such that OPQ is an equilateral triangle, O being the centre of the hyperbola, then the eccentricity e of the hyperbola satisfies

(A) $1 < \mathrm{e} < \frac{2}{\sqrt{3}}$

(B) $\mathrm{e} = \frac{2}{\sqrt{3}}$

(C) $\mathrm{e} = \frac{\sqrt{3}}{2}$

(D) $e > \frac{2}{\sqrt{3}}$

Q.11

Let $\mathrm{P}(a \sec \theta, b \tan \theta)$ and $\mathrm{Q}(\mathrm{a}\sec \varphi, b \tan \varphi)$ where $\theta + \varphi = \pi/2$, be two points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If (h, k) is the point of intersection of the normals at P and Q then K is equal to

(A) $a^2 + b^2/a$

(B) $-\left(a^2 + b^2\right)/a$

(C) $a^2 + b^2/b$

(D) $-\left(a^2 + b^2\right)/b$

Q.12

If the chords of contact of tangents from two points $(-4, 2)$ and $(2, 1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then eccentricity of hyperbola is ?

(A) $\sqrt{2}$

(B) $\sqrt{\frac{3}{2}}$

(C) $\sqrt{\frac{5}{2}}$

(D) $\sqrt{3}$

Q.13

If θ is the angle between the asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with eccentricity e, then $\sec \frac{\theta}{2}$ is equal to

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(A) e

(B) $\frac{e}{2}$

(C) $\frac{e}{3}$

(D) none of these

Q.14

For the hyperbola $\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$, which of the following remains constant when varies?

(A) Eccentricity

(B) Directrix

(C) Abscissae of vertices

(D) Abscissae of foci

Q.15

The equation $\left|\sqrt{x^2+(y-1)^2} - \sqrt{x^2+(y+1)^2}\right| = k$ will represent a hyperbola for

(A) $k \in (0, 2)$

(B) $k \in (0, 1)$

(C) $k \in (1, \infty)$

(D) $k \in R^+$

Q.16

If the normal at P to the rectangular hyperbola $x^2 - y^2 = 4$ meets the axes in G and g and C is the centre of the hyperbola, then

(A) $PG = PC$

(B) $Pg = PC$

(C) $Gg = 2PC$

(D) All of these

Q.17

The circle described on the line joining the foci of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ as a diameter passes through an end of the latus rectum of the parabola $y^2 = 4ax$ the length of the latus rectum of the parabola is

(A) $2\sqrt{5}$ units

(B) 5 units

(C) $4\sqrt{5}$ units

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(D) $5\sqrt{5}$ units

Q.18

$(a\sec\alpha, b\tan\alpha)$ and $(a\sec\beta, b\tan\beta)$ be the ends of a chord of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, passing through the focus $(ae, 0)$ then $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}$ is equal to:

(A) $\frac{1+e}{1-e}$

(B) $\frac{e+1}{e-1}$

(C) $\frac{1-e}{1+e}$

(D) $\frac{e-1}{e+1}$

Q.19

If the latus rectum of a hyperbola subtend an angle of 60° at the other focus, then eccentricity of the hyperbola is

(A) 2

(B) $\frac{\sqrt{3}+1}{2}$

(C) $2\sqrt{3}$

(D) $\sqrt{3}$

Q.20

If a variable line has its intercepts on the coordinate axes e, e', where $\frac{e}{2}, \frac{e'}{2}$ are the eccentricities of a hyperbola and its conjugate hyperbola, then the line always touches the circle $x^2 + y^2 = r^2$, where $r =$

(A) 1

(B) 2

(C) 3

(D) cannot be decided

Q.21

The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide, then the value of b^2 is

Q.22

A tangent drawn to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $P\left(\frac{\pi}{6}\right)$ forms a triangle of area $3a^2$ sq. units with coordinate axes. Square of the eccentricity of the hyperbola is equal to

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Q.23

A variable chord $PQ, x\cos\theta + y\sin\theta = p$ of the hyperbola $\frac{x^2}{8} - \frac{y^2}{16} = 1$ angle at the origin. If this chord always touches a circle, then find its radius.

Q.24

Tangent at the point $(2\sqrt{2}, 3)$ to the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$ meets its asymptotes at A and B, then area of the triangle OAB, O being the origin is

Q.25

If the tangent and normal to the hyperbola $x^2 - y^2 = 4$ at a point cut off intercepts a_1 and a_2 respectively on the x-axis, and b_1 and b_2 respectively on the y-axis, then the value of $a_1a_2 + b_1b_2$ is

Q.26

If P is the length of the perpendicular from a focus upon the tangent at any point P on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and r is the distance of P from that focus, then $\frac{b^2}{p^2} - \frac{2a}{r} =$

Q.27

The area of triangle formed by the tangents from point (3, 2) to hyperbola $x^2 - 9y^2 = 9$ and the chord of contact w.r.t. point $(3, 2)$ is

Q.28

If $(5, 12)$ and $(24, 7)$ are the foci of a hyperbola passing though the origin then the eccentricity of the hyperbola is (mark answer to nearest integer)

Q.29

Let a and b respectively be the semi-transverse and semi-conjugate axes of a hyperbola whose eccentricity satisfies the equation $9e^2 - 18e + 5 = 0$. If $S(5, 0)$ is a focus and $5x = 9$ is corresponding directrix of this hyperbola, then $|a^2 - b^2|$ is equal to:

Q.30

If the asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{36} = 1$ are perpendicular to the asymptotes of the hyperbola $\frac{x^2}{49} - \frac{y^2}{b^2} = 1$ then

Answers & Solutions

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Q.1 Answer:

$x = 4, y = 3$

Solution:

$xy - 3x - 4y + 8 = 0$

Asymptotes are $xy - 3x - 4y + c = 0$ **such that it's a pair of straight lines i.e.,**

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \Rightarrow c = 12$

$x(y-3) - 4(y-3) = 0$

$x = 4$ and $y = 3$

Q.2 Answer:

$(2, -1)$ or $(-2, 1)$

Solution:

Clearly tangents at $(2, -1)$ **or** $(-2, 1)$ **are parallel to** $2x + y + 8 = 0$

Q.3 Answer:

all of these

Solution:

Since $16x^2 - 32x = 16(x-1)^2 - 16$ **and** $3y^2 + 12y = 3(y+2)^2 - 12$**, the given equation can be written** $16(x-1)^2 - 16 - 3(y+2)^2 + 12 - 44 = 0$

$\Rightarrow \quad 16(x-1)^2 - 3(y+2)^2 - 48 = 0$

$\Rightarrow \quad \frac{(x-1)^2}{3} - \frac{(y+2)^2}{16} = 1$

Comparing it to the standard equation of a hyperbola centered at the origin, $\frac{x^2}{a^2} = \frac{y^2}{b^2} = 1$**, we see that the given curve is a hyperbola with centre at** $(1, -2)$**, and for which** $a = \sqrt{3}$ **and** $b = 4$**. Thus the length of the transverse axis is** $2a = 2\sqrt{3}$**; the length of the conjugate axis is** $2b = 8$**; and the eccentricity is**

$e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{16}{3}} = \sqrt{\frac{19}{3}}$

Q.4 Answer:

$4x^2 + 4y^2 - 3 = 0$

Solution:

The hyperbola $4y^2 = x^2 - 1$ **can be rewritten as** $\frac{x^2}{1} - \frac{y^2}{1/4} = 1$ **will be**

$x^2 + y^2 = 1 - \frac{1}{4}$

Q.5 Answer:

a^2

Solution:

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Let $P(x_1, y_1)$ be a point on the hyperbola. Then the coordinates of N are $(x, 0)$.
The equation of the tangent at (x_1, y_1) is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$
This meets x-axis at $T\left(\frac{a^2}{x_1}, 0\right)$
$\therefore OT.ON = \frac{a^2}{x_1} \times x_1 = a^2$

Q.6 Answer:

$y^2 + b^2 = c^2\left(x^2 - a^2\right)$

Solution:

Any tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $y = mx + \sqrt{a^2m^2 - b^2}$
This passes through (x_1, y_1) if $(y_1 - mx_1)^2 = a^2m^2 - b^2$

$$\Rightarrow \quad \left(x_1^2 - a^2\right)m^2 - 2x_1y_1m + y_1^2 + b^2 = 0$$

$$c^2 = m_1m_2 = \frac{y_1^2 + b^2}{x_1^2 - a^2} \quad \therefore \quad y_1^2 + b^2 = c^2\left(x_1^2 - a^2\right)$$

Locus is $y^2 + b^2 = c^2\left(x^2 - a^2\right)$

Q.7 Answer:

transverse axis is y-axis

Solution:

Since coefficient of x^2 is negative, therefore
The hyperbola is of the form $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$, intersect y-axis in real points So transverse axis is y-axis.

Q.8 Answer:

$4x^2 - 3y^2 - 36 = 0$

Solution:

We have, $a = 3$ and $\frac{b^2}{a} = 4$
Hence, the equation of the hyperbola is $\frac{x^2}{9} - \frac{y^2}{12} = 1 \quad \Rightarrow 4x^2 - 3y^2 = 36$

Q.9 Answer:

$x^2\operatorname{cosec}^2\theta - y^2\sec^2\theta = 1$

Solution:

Foci of the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ is $(\pm 1, 0)$
Let the equation of the hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
Such that $2a = 2\sin\theta \Rightarrow a = \sin\theta$.
and $a^2 + b^2 = a^2e^2 = 1 \Rightarrow b = \cos\theta$.
So the required equation is

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$\frac{x^2}{\sin^2\theta} - \frac{y^2}{\cos^2\theta} = 1$
or $x^2 \operatorname{cosec}^2\theta - y^2 \sec^2\theta = 1$

Q.10 Answer:

$e > \frac{2}{\sqrt{3}}$

Solution:

Let the hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **and any double ordinate** PQ **be (a** $\sec\theta, b\tan\theta$ **), (a** $\sec\theta, -b\tan\theta$ **) and** O **is centre** $(0,0)$.

$\triangle OPQ$ **being equilateral.**

Thus $\tan 30^\circ = \frac{b\tan\theta}{a\sec\theta} \Rightarrow 3 \cdot \frac{b^2}{a^2} = \csc^2\theta \Rightarrow 3\left(e^2-1\right) = \csc^2\theta$

Now, $\csc^2\theta \geq 1$

Therefore, $3\left(e^2-1\right) \geq 1 \Rightarrow e^2 \geq \frac{4}{3}$

$\Rightarrow e > 2/\sqrt{3}$

Q.11 Answer:

$-\left(a^2+b^2\right)/b$

Solution:

The point of intersection of the normals at P **and** Q **are given as**

$$\left\{\frac{-\left(\frac{a^2+b^2}{a}\right)\cos\frac{\theta-\phi}{2}\cos\theta\cos\phi}{\cos\left(\frac{\theta+\phi}{2}\right)}, -\left(\frac{a^2+b^2}{b}\right)\tan\left(\frac{\theta+\phi}{2}\right)\tan\theta\tan\phi\right\}$$

Since $\theta + \varphi = \pi/2$

$\therefore \quad k = -\frac{a^2+b^2}{b} \cdot \tan\frac{\pi}{4} \cdot \tan\theta\cot\theta = -\frac{a^2+b^2}{b}$

Q.12 Answer:

$\sqrt{\frac{3}{2}}$

Solution:

Chords of contact are $\frac{-4x}{a^2} - \frac{2y}{b^2} = 1$ **and** $\frac{2x}{a^2} - \frac{y}{b^2} = 1$ **These are at right angles**

So, $\frac{-4b^2}{2a^2} \cdot \frac{2b^2}{a^2} = -1$

$\frac{b^4}{a^4} = \frac{1}{4} \Rightarrow \frac{b^2}{a^2} = \frac{1}{2}$

Now, $\frac{b^2}{a^2} = e^2 - 1 \quad \Rightarrow \quad e^2 = \frac{3}{2} \Rightarrow e = \sqrt{\frac{3}{2}}$

Q.13 Answer:

e

Solution:

Equation of asymptotes of the given hyperbola are $\frac{x}{a} + \frac{y}{b} = 0$ **and** $\frac{x}{a} - \frac{y}{b} = 0$

Slope of asymptote $= \frac{b}{a} = \tan\alpha$ (say)

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If θ is the angle between the two asymptotes, then

$$\tan\theta = \frac{\frac{b}{a}+\frac{b}{a}}{1-\frac{b^2}{a^2}} = \frac{\tan\alpha+\tan\alpha}{1-\tan^2\alpha} = \frac{2\tan\alpha}{1-\tan^2\alpha} = \tan 2\alpha$$

$$\therefore \quad \frac{\theta}{2} = \alpha \quad \Rightarrow \quad \tan\frac{\theta}{2} = \tan\alpha = \frac{b}{a}$$

$$\therefore \quad \sec\frac{\theta}{2} = \sqrt{1+\tan^2\frac{\theta}{2}} = \sqrt{1+\frac{b^2}{a^2}} = e$$

Q.14 Answer:

Abscissae of foci

Solution:

$$\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$$

$$\therefore b^2 = a^2\left(e^2-1\right) \Rightarrow \sin^2\alpha = \cos^2\alpha\left(e^2-1\right)$$

$$\Rightarrow \quad e^2 = \tan^2\alpha + 1 = \sec^2\alpha \Rightarrow e = \sec\alpha$$

Directrix: $x = \pm\frac{a}{e} = \pm\cos^2\alpha$

Abscissae of vertices $= \pm a = \pm\cos\alpha$

Abscissae of foci $= \pm ae = \pm\cos\alpha\cdot\sec\alpha = \pm 1$

Correct choice: (d)

Q.15 Answer:

$k \in (0,2)$

Solution:

$\left|\sqrt{[x^2+(y+1)^2]} - \sqrt{[x^2+(y-1)^2]}\right| = k$ **which is equivalent to** $|S_1P - S_2P| =$ **constant where** $S_1 = (0,1), S_2 = (0,-1)$

If above equation represents hyperbola then $k = 2a$ **where** $2a$ **is transverse axis and e is eccentricity.**

also $2ae = S_1\,S_2 = 2$

$\therefore$ **Dividing** $(k/2) = (2a/2ae)$

$\therefore (k/2) = (1/e)$

i.e. $e = (2/k)$

For hyperbola $e > 1$

$\Rightarrow e = (2/k) > 7$

$\Rightarrow 2 > k$

i.e. $k < 2$

i.e. k must be a positive quantity

$\Rightarrow k \in (0,2)$

Q.16 Answer:

All of these

Solution:

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Let $P(x_1, y_1)$ be any point on the hyperbola $x^2 - y^2 = 4$, then equation of the normal at P is

$y - y_1 = -\frac{y_1}{x_1}(x - x_1) \Rightarrow x_1 y + y_1 x = 2x_1 y_1$

Then coordinates of G are $(2x_1, 0)$ and of g are $(0, 2y_1)$

So that $PG = \sqrt{(2x_1 - x_1)^2 + y_1^2} = \sqrt{x_1^2 + y_1^2} = PC$

$Pg = \sqrt{x_1^2 + (2y_1 - y_1)^2} = \sqrt{x_1^2 + y_1^2} = PC$

and $\quad Gg = \sqrt{(2x_1)^2 + (2y_1)^2} = 2\sqrt{x_1^2 + y_1^2} = 2PC$

Q.17 Answer:

$4\sqrt{5}$ units

Solution:

Eccentricity of the hyperbola is given by $e = \sqrt{\frac{16+9}{16}} = \frac{5}{4}$

Coordinates of the foci are $\left(\pm 4 \times \frac{5}{4}, 0\right) = (\pm 5, 0)$.

So the circle described on the line joining the foci is $x^2 + y^2 = 25$

If it passes through an end $(a, \pm 2a)$ of the latus rectum of the parabola $y^2 = 4ax$ then $a^2 + 4a^2 = 25 \Rightarrow a^2 = 5 \Rightarrow a = \sqrt{5}$.

Length of the latus rectum $= 4a = 4\sqrt{5}$.

Q.18 Answer:

$\frac{1-e}{1+e}$

Solution:

Equation of the chord joining the given points is

$$\begin{vmatrix} x & y & 1 \\ a\sec\alpha & b\tan\alpha & 1 \\ a\sec\beta & b\tan\beta & 1 \end{vmatrix} = 0$$

$bx(\tan\alpha - \tan\beta) - ay(\sec\alpha - \sec\beta) + ab(\sec\alpha\tan\beta - \sec\beta \tan\alpha) = 0$

$\Rightarrow bx\sin(\alpha - \beta) - ay(\cos\beta - \cos\alpha) + ab(\sin\beta - \sin\alpha) = 0$

$\Rightarrow bx\cos\frac{\alpha-\beta}{2} - ay\sin\frac{\alpha+\beta}{2} - ab\cos\frac{\alpha+\beta}{2} = 0$

$\Rightarrow \frac{x}{a}\cos\frac{\alpha-\beta}{2} - \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha+\beta}{2}$

If it passes through $(0e, 0) e\cos\frac{\alpha-\beta}{2} = \cos\frac{\alpha+\beta}{2}$

$\Rightarrow \frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}} = e$

$\Rightarrow \frac{\cos\frac{\alpha+\beta}{2} - \cos\frac{\alpha-\beta}{2}}{\cos\frac{\alpha-\beta}{2} + \cos\frac{\alpha-\beta}{2}} = \frac{e-1}{e+1}$

$\Rightarrow -\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{e-1}{e+1}$

$\Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{1-e}{1+e}$

Q.19 Answer:

$\sqrt{3}$

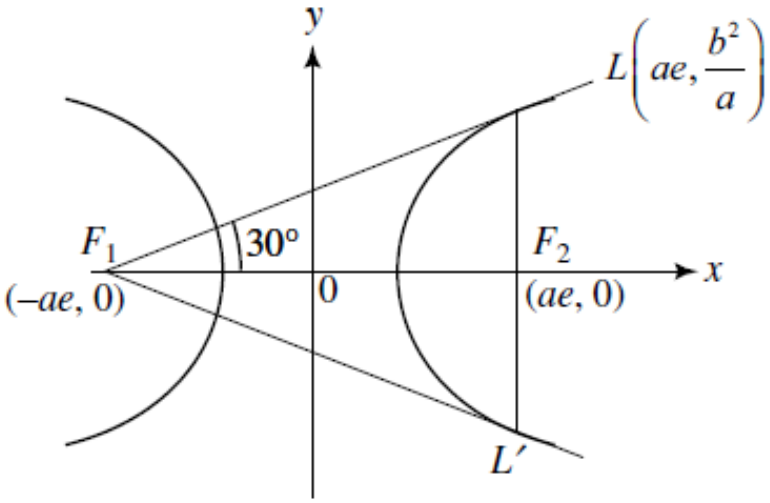
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Solution:

Let the equation of the hyperbola be

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$F_1(-ae, 0), F_2(ae, 0)$ **be the foci,** e **being the eccentricity. Let** LF_2L'**, the latus rectum subtend an angle of** 60° **at** F_1**, then slope of** LF_1 **is** $\tan 30^\circ$**.**



So $\tan 30^\circ = \frac{b^2/a}{2ae}$

$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{a^2(e^2-1)}{2a^2e} = \frac{e^2-1}{2e}$

$\Rightarrow \quad e^2 - 1 = \frac{2}{\sqrt{3}}e \Rightarrow \left(e - \frac{1}{\sqrt{3}}\right)^2 = 1 + \frac{1}{3}$

$\Rightarrow \quad e - \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}} \Rightarrow e = \sqrt{3}$

Q.20 Answer:

2

Solution:

Since $\frac{e}{2}$ **and** $\frac{e'}{2}$ **are eccentricities of a hyperbola and its conjugate hyperbola, therefore**

$\frac{4}{e^2} + \frac{4}{e'^2} = 1$

$\Rightarrow \quad 4 = \frac{e^2e'^2}{e'^2+e'^2}$

The line passing through the points $(e, 0)$ **and** $(0, e')$ **is**

$e'x + ey - ee' = 0$

It is tangent to the circle $x^2 + y^2 = r^2$

Hence, $\frac{ee'}{\sqrt{e^2+e'^2}} = r$

$\Rightarrow \quad r^2 = \frac{e^2e'^2}{e^2+e'^2} = 4$

$\Rightarrow \quad r = 2$

Q.21 Answer:

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Solution:

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For the given ellipse, $a^2 = 16, b^2 = b^2$.

$\therefore \quad e = 1 - \frac{b^2}{a^2} \Rightarrow e = \frac{\sqrt{16-b^2}}{4}$

Thus the foci of the ellipse are $(\pm ae, 0)$ **i.e.** $\left(\pm\sqrt{(16-b^2)}, 0\right)$. **Re-arranging the equation of the hyperbola, we get**

$\left[x^2\left(\frac{12}{5}\right)^2\right] - \left[y^2\left(\frac{9}{5}\right)^2\right] = 1$

Comparing this with standard equation, we have

$a = \frac{12}{5}, b = \frac{9}{5}$ and the centre $(0, 0)$

$e^2 = 1 + \left(\frac{b^2}{a^2}\right) \quad \Rightarrow \quad e = \frac{5}{4}$

Thus the foci of the hyperbola are $(\pm ae, 0)$ **i.e.** $(\pm 3, 0)$

Since the foci of ellipse and hyperbola coincide, therefore $\sqrt{(16-b^2)} = 3 \quad \Rightarrow b^2 = 7$

Q.22 Answer:

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Solution:

Any arbitrary tangent is $\frac{x}{a}\sec\theta - \frac{y}{b}\tan\theta = 1$

If it meets the x**-axis and** y**-axis at the points** A **and** B **respectively,**

then $A \equiv (a\cos\theta, 0), B \equiv (0, b\cot\theta)$

$\Delta_{OAB} = \frac{1}{2}ab\frac{\cos^2\theta}{\sin\theta} = \frac{ab}{2}\cdot\frac{3}{4}\cdot\frac{2}{1} = \frac{3ab}{4} = 3a^2$ **(given)**

$\Rightarrow \quad b = 4a \Rightarrow 16a^2 = a^2\left(e^2 - 1\right) \Rightarrow e = \sqrt{17}$

Q.23 Answer:

—

Solution:

Combined equation of lines joining the points of intersection of the chord with origin is

$\frac{x^2}{8} - \frac{y^2}{16} = \left(\frac{x\cos\theta + y\sin\theta}{p}\right)^2$

$\Rightarrow \quad x^2\left(\frac{p^2}{8} - \cos^2\theta\right) - y^2\left(\frac{p^2}{16} + \sin^2\theta\right) - xy\sin 2\theta = 0$

These lines are mutually perpendicular provided $\frac{p^2}{8} - \cos^2\theta - \frac{p^2}{16} - \sin^2\theta = 0$

$\Rightarrow \quad p^2 = 16$

Thus chord becomes $x\cos\theta + y\sin\theta = 4$, **which is clearly a tangent to the circle having centre at origin and radius equal to** 4.

Q.24 Answer:

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Solution:

Since area of the triangle formed by tangent at any point lying on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **and its asymptotes is always constant and is equal to** ab **Therefore, required area is** $2 \times 3 = 6$ **sq. units**

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Q.25 Answer:

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Solution:

Let the coordinates of a point P on the hyperbola be (x_1, y_1).

Equation of the tangent at P to the hyperbola is

$xx_1 - yy_1 = 4 \Rightarrow a_1 = \frac{4}{x_1}, b_1 = -\frac{4}{y_1}$

Equation of the normal is

$y - y_1 = -\frac{y_1}{x_1}(x - x_1)$

$y_1 x + x_1 y = 2x_1 y_1$

$\Rightarrow a_2 = 2x_1, b_2 = 2y_1$

So that $a_1 a_2 + b_1 b_2 = 0$

Q.26 Answer:

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Solution:

Equation of the tangent at $P(a\sec\theta, b\tan\theta)$ to the hyperbola is

$\frac{x}{a}\sec\theta - \frac{y}{b}\tan\theta = 1$

Length of the perpendicular from the focus $S(ae, 0)$ to this tangent is

$$P = \left|\frac{e\sec\theta - 1}{\sqrt{\frac{\sec^2\theta}{a^2} + \frac{\tan^2\theta}{b^2}}|}\right| = \frac{ab(e\sec\theta - 1)}{\sqrt{b^2\sec^2\theta + a^2\tan^2\theta}}$$

$$= \frac{ab(e\sec\theta - 1)}{\sqrt{a^2(e^2-1)\sec^2\theta + a^2\tan^2\theta}}$$

$$= \frac{ab(e\sec\theta - 1)}{a\sqrt{e^2\sec^2\theta - 1}} = b\sqrt{\frac{e\sec\theta - 1}{e\sec\theta + 1}}$$

$$\Rightarrow \quad \frac{b^2}{p^2} = \frac{e\sec\theta + 1}{e\sec\theta - 1}$$

Now $r^2 = (ae - a\sec\theta)^2 + b^2\tan^2\theta$

$$= a^2\left[e^2 - 2e\sec\theta + \sec^2\theta\right] + a^2\left(e^2 - 1\right)\tan^2\theta$$

$$= a^2\left[e^2\sec^2\theta - 2e\sec\theta + 1\right]$$

$$= a^2(e\sec\theta - 1)^2$$

$$\Rightarrow \quad \frac{r^2}{a^2} = (e\sec\theta - 1)^2 \Rightarrow \frac{a}{r} = \frac{1}{e\sec\theta - 1}$$

Q.27 Answer:

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Solution:

Hyperbola is $x^2 - 9y^2 = 9$ or $\frac{x^2}{9} - \frac{y^2}{1} = 1$

Equation of tangent is $y = mx \pm \sqrt{a^2m^2 - b^2}$ (1)

It passes through $(3, 2)$

$\Rightarrow \quad 2 = 3m \pm \sqrt{9m^2 - 1}$

or $4 + 9m^2 - 12m = 9m^2 - 1$

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Solving we get values of m as $m_1 = \frac{5}{12}$ and $m_2 = \infty$ Equation of tangent (1) for $m_1 = \frac{5}{12}$

$y = \frac{5}{12}x \pm \sqrt{9\left(\frac{5}{12}\right)^2 - 1}$

or $y = \frac{5}{12}x \pm \frac{3}{4}$

on taking (-)ve sign point $P(3, 2)$ does not satisfy the equation of tangent therefore rejecting (-)ve sign. Hence equation of tangent is $y = \frac{5x}{12} + \frac{3}{4}$ (2)

now equation of tangent (1) for $m_2 = \infty$ is $x \pm 3 = 0$ (3)

rejecting $(+)$ sign (since taking $(+)$ sign point $P(3, 2)$ does not satisfy this equation.)

Hence equation of tangent is $x - 3 = 0$ (4)

Now equation of chord of contact w.r.t. point $P(3, 2)$ is $T = 0$

or $3x - 18y = 9$

or $x - 6y = 3$

Solving (2) and (4); $x = -5, y = -\frac{4}{3}$

Solving (3) and (4); $x = 3, y = 0$

Now vertices of triangle are $(3, 2), (3, 0), (-5, -4/3)$

$$\therefore \text{ Area } = \frac{1}{2}\begin{vmatrix} 3 & 2 & 1 \\ 3 & 0 & 1 \\ -5 & -\frac{4}{3} & 1 \end{vmatrix}$$

$$= \frac{1}{2} \times \left|3\left(\frac{4}{3}\right) - 2(3+5) + 1(-4)\right|$$

$$= \frac{1}{2}|4 - 16 - 4|$$

$$= 8 \text{ sq. units}$$

Q.28 Answer:

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Solution:

Let $S(5, 12)$ and $S'(24, 7)$ be the two foci and $P(0, 0)$ be a point on the conic

then

$$SP = \sqrt{25 + 144} = \sqrt{169} = 13$$

$$S'P = \sqrt{(24)^2 + 7^2} = \sqrt{625} = 25$$

$$SS' = \sqrt{(24-5)^2 + (7-12)^2} = \sqrt{19^2 + 5^2}$$

$$= \sqrt{386}$$

since the conic is a hyperbola, $S'P - SP = 2a$, the length of transverse axis and $SS' = 2ae, e$ being the eccentricity.

$\Rightarrow \quad e = \frac{SS'}{S'P - SP} = \frac{\sqrt{386}}{12}$

Q.29 Answer:

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Solution:

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As $9e^2 - 18e + 5 = 0$

$\Rightarrow 9e^2 - 15e - 3e + 5 = 0$

$\Rightarrow 3e(3e - 5) - (3e - 5) = 0$

$\Rightarrow (3e - 1)(3e - 5) = 0 \Rightarrow e = 1/3, 5/3$

As $e > 1, e = 5/3$

We have $ae = 5 \Rightarrow a = 3$

Also, $a^2 - b^2 = a^2 - a^2\left(e^2 - 1\right)$

$= 9(2 - 25/9) = -7$

Q.30 Answer:

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Solution:

Asymptotes to the two hyperbola are

$\frac{x}{a} \pm \frac{y}{6} = 0$ and $\frac{x}{7} \pm \frac{y}{b} = 0$

If they are at right angles then $\left(\pm\frac{6}{a}\right)\left(\pm\frac{b}{7}\right) = -1$

$\Rightarrow 7a \pm 6b = 0$

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